

XML GUIDE FOR DUALSPHYSICS

**FLEXIBLE FLUID-STRUCTURE
INTERACTION**

SPECIAL: FLEXSTRUCS



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flexstrucs: Flexible fluid-structure interaction

A **flexible fluid-structure interaction** formulation is available from DualSPHysics v5.2. The approach is based on a Total Lagrangian SPH formulation for the dynamics of the flexible structure, and the fluid-structure interaction is handled via the existing boundary conditions within DualSPHysics.

There are three main developments implemented within the structural dynamics solver: 1) a **Total Lagrangian formulation**, which relies on representing all relevant quantities and operators with respect to an initial (material) reference frame, **to eliminate tensile instabilities**; 2) a **kernel correction** scheme for the flexible structures **to retain linear consistency** at the edges of the geometry; 3) an **hourglass (zero-energy) mode suppression** scheme to **reduce the effect of instabilities related to rank-deficiency** of the collocated SPH discretisation.

The **interaction at the fluid-structure interface** is handled automatically by the existing **boundary conditions** within DualSPHysics. In this way, the fluid sees the flexible structure as a normal moving boundary.

For more information, please see **O'Connor, J. and Rogers, B.D. *Journal of Fluids and Structures*, 2021.**

```
<commands>
  <mainlist>
    <setdrawmode mode="full" />
    <!--Clamp-->
    <setmkbound mk="0" />
    <drawsphere radius="0.05">
      <point x="0.2" y="0.0" z="0.2" />
    </drawsphere>
    <!--Flexible Structure-->
    <setmkbound mk="1" />
    <drawbox>
      <boxfill>solid</boxfill>
      <point x="0.2" y="-0.01" z="0.19" />
      <size x="0.4" y="0.02" z="0.02" />
    </drawbox>
  </mainlist>
</commands>
</geometry>
<motion>
  <objreal ref="1">
    <begin mov="1" start="0" />
    <mvnull id="1" />
  </objreal>
</motion>
</casedef>
<execution>
  <special>
    <flexstrucs>
      <flexstrucbody mkbound="1" mkclamp="0">
        <density value="1000.0" comment="Mass density" units_comment="kg/m^3" />
        <youngmod value="1.4e6" comment="Young's Modulus" units_comment="Pa" />
        <poissratio value="0.4" comment="Poisson ratio" />
        <constitmodel value="1" comment="Constitutive model 1:Plane Strain (2D),>
        <hgfactor value="0.0" comment="Hourglass correction factor: keep as low .>
      </flexstrucbody>
    </flexstrucs>
  </special>
```

Geometry definition

Tag as moveable object

Flexible structure definition

There are three main steps to setting up a case involving flexible fluid-structure interaction:

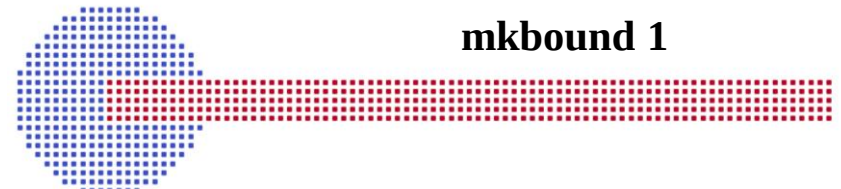
- 1) Define the geometry of the flexible structure and where it will be clamped.
- 2) Tag the flexible structure as a moveable object.
- 3) Tag the flexible structure and its clamp and define the material and numerical properties.

Geometry Definition

```
<mainlist>
  <setdrawmode mode="full" />
  <!--Clamp-->
  <setmkbound mk="0" />
  <drawsphere radius="0.05">
    <point x="0.2" y="0.0" z="0.2" />
  </drawsphere>
  <!--Flexible Structure-->
  <setmkbound mk="1" />
  <drawbox>
    <boxfill>solid</boxfill>
    <point x="0.2" y="-0.01" z="0.19" />
    <size x="0.4" y="0.02" z="0.02" />
  </drawbox>
</mainlist>
```

It is necessary to define the geometry of the clamp first, then embed the flexible structure within the clamp.

mkbound 0



Tag as Moveable Object

```
<motion>
  <objreal ref="1">
    <begin mov="1" start="0" />
    <mvnull id="1" />
  </objreal>
</motion>
```

The flexible structure is tagged as a moveable object via the **<motion/>** block by setting the **<mvnull/>** label. This tells DualSPHysics that the motion is to be computed during runtime.

Flexible Structure Definition

```
<special>
  <flexstrucs>
    <flexstrucbody mkbound="1" mkclamp="0">
      <density value="1000.0" comment="Mass density" units_comment="kg/m^3" />
      <youngmod value="1.4e6" comment="Young's Modulus" units_comment="Pa" />
      <poissratio value="0.4" comment="Poisson ratio" />
      <constitmodel value="1" comment="Constitutive model 1:Plane Strain (2D),
      <hgfactor value="0.0" comment="Hourglass correction factor: keep as low
    </flexstrucbody>
  </flexstrucs>
</special>
```

mkbound and **mkclamp** are the mkbound numbers for the flexible structure and clamp, respectively.

density is the mass density (kg/m^3) of the flexible structure, **youngmod** is the Young's modulus (Pa), and **poissratio** is the Poisson ratio.

constitmodel is the constitutive model to be used. This defines the stress-strain relationship for the flexible structure. There are two available in 2-D (plane strain and plane stress) and one for 3-D (St. Venant-Kirchhoff). See the end of this document for more details.

hgfactor is the hourglass correction factor to use in the zero-energy mode suppression scheme. Note that non-zero values will increase the effective stiffness of the flexible structure. Therefore, it is recommended to set this to zero first and only increase it if zero-energy instabilities appear. If required, a value of 0.1 has been found to work effectively with negligible impact on the effective stiffness.

Geometry Definition

```
<commands>
  <mainlist>
    <setdrawmode mode="full" />
    <setmkfluid mk="0" />
    <drawbox>
      <boxfill>solid</boxfill>
      <point x="0.0" y="-0.01" z="0.0" />
      <size x="0.2" y="0.02" z="0.4" />
    </drawbox>
    <setmkbound mk="0" />
    <drawbox>
      <boxfill>solid</boxfill>
      <point x="0.2" y="-0.01" z="0.0" />
      <size x="0.008" y="0.02" z="0.5" />
    </drawbox>
    <setmkbound mk="1" />
    <drawbox>
      <boxfill>solid</boxfill>
      <point x="-0.008" y="-0.01" z="-0.016" />
      <size x="0.816" y="0.02" z="0.016" />
    </drawbox>
    <setmkbound mk="2" />
    <drawbox>
      <boxfill>solid</boxfill>
      <point x="0.596" y="-0.01" z="-0.008" />
      <size x="0.008" y="0.02" z="0.098" />
    </drawbox>
    <setmkbound mk="3" />
    <drawbox>
      <boxfill>solid</boxfill>
      <point x="-0.008" y="-0.01" z="-0.016" />
      <size x="0.008" y="0.02" z="0.616" />
    </drawbox>
    <drawbox>
      <boxfill>solid</boxfill>
      <point x="0.8" y="-0.01" z="-0.016" />
      <size x="0.008" y="0.02" z="0.616" />
    </drawbox>
  </mainlist>
</commands>
```

mkbound 3

mkbound 0

mkfluid 0

mkbound 3

mkbound 2

mkbound 1

Tag as Moveable Object

```
<motion>
  <objreal ref="0">
    <begin mov="1" start="0.0" finish="0.12" />
    <mvfile id="1" duration="0.12">
      <file name="DambreakGateTimeSeries.csv" fields="2" fieldtime="0" fieldz="1" />
    </mvfile>
  </objreal>
  <objreal ref="2">
    <begin mov="1" start="0" />
    <mvnull id="1" />
  </objreal>
</motion>
```

As well as tagging the flexible structure as a moveable object, a realistic gate motion is also included to initiate the dam break.

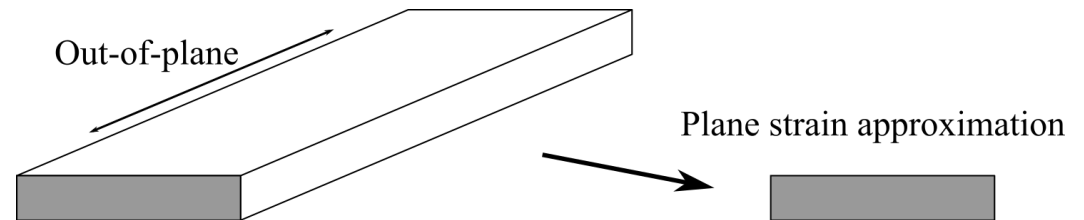
Flexible Structure Definition

```
<special>
  <flexstrucs>
    <flexstrucbody mkbound="2" mkclamp="1">
      <density value="1161.54" comment="Mass density" units_comment="kg/m^3" />
      <youngmod value="8.75e5" comment="Young's Modulus" units_comment="Pa" />
      <poissratio value="0.49" comment="Poisson ratio" />
      <constitmodel value="1" comment="Constitutive model 1:Plane Strain (2D),
      <hgfactor value="0.1" comment="Hourglass correction factor: keep as low a
    </flexstrucbody>
  </flexstrucs>
</special>
```

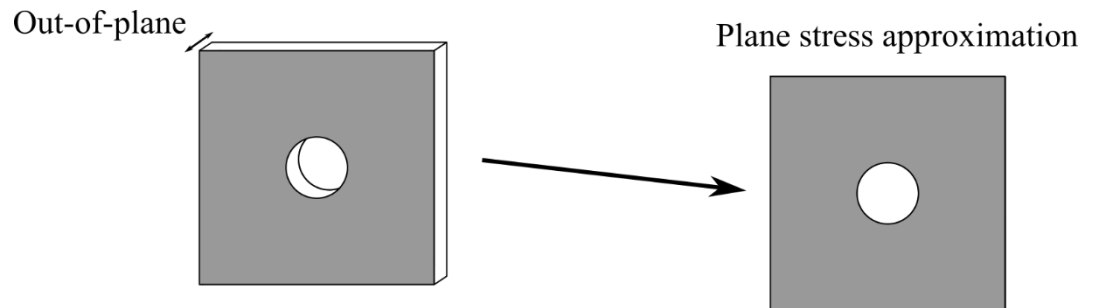
Constitutive Model

There are three possible options for **constitmodel**: 1 (plane strain), 2 (plane stress), and 3 (St. Venant-Kirchhoff).

Plane strain is only valid for 2-D problems and models the third (out-of-plane) dimension by assuming there is **zero strain in the third dimension**. This is suitable for problems that are **very thick in the out-of-plane direction**.



Plane stress is only valid for 2-D problems and models the third (out-of-plane) dimension by assuming there is **zero stress in the third dimension**. This is suitable for problems that are **very thin in the out-of-plane direction**.



The hyperelastic **St. Venant-Kirchhoff** constitutive model is the only available 3-D model and is an extension of the **linear elastic** model to the **geometrically nonlinear regime**.

Setting Up mDBC on a Flexible Structure

Setting up mDBC on a flexible structure is the same as any other fixed or moving boundary and just requires defining the surface normals. See **XML_GUIDE_MDBC.pdf** for more details.

However, for now, there are some restrictions:

- 1) mDBC can only be defined on a flexible structure if using the Symplectic time integration scheme (**StepAlgorithm=2**).
- 2) mDBC can only be defined on a flexible structure if using either no-slip (**SlipMode=2**) or free-slip (**SlipMode=3**).

If these conditions are not met, then DualSPHysics will raise an exception.

Dealing with Particle Penetration

For some violent flows, mDBC is known to exhibit some particle penetration through the boundary. This causes problems for the fluid-structure coupling. Therefore, for certain problems, it may be necessary to turn on the no-penetration scheme (**NoPenetration=1**).